



**AnimoNexus: A Web-Based Repository of Approved Research
Papers for University Research Office at De La Salle University -
Dasmariñas**

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APPROVAL SHEET

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CHAPTER I

INTRODUCTION

1.1 Introduction

The rise of institutional repositories (IRs) has fundamentally transformed how academic research is preserved and disseminated. These platforms serve as centralized hubs for collecting and sharing the scholarly output of universities, significantly enhancing the visibility of research work within the academic community. Recent studies underscore the vital role of IRs in improving access to research, which in turn increases citation rates and overall scholarly impact (Büchler & Hoyer, 2020; Peroni, 2021). By offering a systematic approach to managing academic contributions, institutional repositories not only facilitate knowledge sharing but also support compliance with emerging open access policies, which are increasingly important in higher education (Meyer & Rowlands, 2021).

At De La Salle University - Dasmariñas, the University Research Office (URO) oversees the quality and integrity of research outputs. However, the absence of a centralized platform for approved research papers has created challenges for students and faculty seeking access to these vital academic resources. The establishment of **AnimoNexus**, a centralized online repository, aims to address these challenges by ensuring that only URO-approved research



papers are accessible to university affiliates, thereby promoting a culture of academic excellence and collaboration within the institution. By providing an organized and efficient means of accessing scholarly work, AnimoNexus aligns with the institution's goals of enhancing research engagement and facilitating a vibrant academic community.

1.2 Project Context

In today's rapidly evolving academic landscape, effective management and dissemination of research are critical to advancing knowledge and fostering innovation. De La Salle University - Dasmariñas recognizes the importance of supporting its faculty and students in their research endeavors while ensuring that academic contributions are easily accessible and preserved for future generations. However, the current decentralized system of storing research papers presents significant challenges that hinder collaboration and knowledge sharing within the university community.

The University Research Office (URO) plays a pivotal role in overseeing research activities and ensuring that all research outputs meet institutional standards. Yet, without a centralized repository, there is an increased risk of valuable research being overlooked or lost in the absence of a systematic archiving process. Furthermore, the lack of a unified platform limits the visibility of DLSU-Dasmariñas' academic work, reducing the potential impact of its research efforts both within and beyond the university.



AnimoNexus aims to address these challenges by providing a dedicated, web-based repository where only URO-approved research papers are curated and made accessible exclusively to individuals within the campus. This project will not only streamline the storage and retrieval of academic papers but will also promote a culture of knowledge-sharing and collaboration among students and faculty. By centralizing research efforts, **AnimoNexus** will enhance the university's capacity to preserve and showcase its academic contributions, reinforcing its commitment to fostering a vibrant academic community that values innovation and continuous learning.

The development of **AnimoNexus** represents a crucial step toward modernizing the management of research outputs at De La Salle University - Dasmariñas, ensuring that the intellectual contributions of its community are recognized, accessible, and preserved for future academic pursuits.

1.3 Purpose and Description

The purpose of this research project, **AnimoNexus**, is to create a centralized, web-based repository for research papers at De La Salle University - Dasmariñas, managed exclusively by the University Research Office (URO). This platform addresses critical challenges in accessibility, searchability, and preservation by providing a structured, user-friendly digital archive that serves as a comprehensive resource for the university community. **AnimoNexus** will



exclusively host research papers that have been approved by the URO, ensuring that only high-quality, validated academic contributions are accessible to users within the campus.

Through this repository, the URO will curate and upload approved research papers, systematically organizing them for easy viewing by students, faculty, and staff at De La Salle University - Dasmariñas. By centralizing these resources, **AnimoNexus** significantly enhances research efficiency, allowing users to quickly locate and reference relevant works without the difficulties associated with scattered documents across multiple locations. The platform's advanced search functionality will enable users to filter results by various criteria, such as author, date, or keywords, streamlining the research process and encouraging collaboration among the university community.

Furthermore, **AnimoNexus** plays a vital role in preserving the university's academic contributions through long-term storage, safeguarding valuable research output for future generations. This repository not only increases the visibility of DLSU-Dasmariñas' research but also underscores the institution's commitment to fostering an environment of knowledge-sharing and collaboration. By promoting easy access to approved academic materials for individuals within the campus, **AnimoNexus** supports the university's mission of advancing education and



research, ultimately enhancing the academic experience for all stakeholders involved.

1.4 Statement of the Problem

The University Research Office (URO) at De La Salle University - Dasmariñas currently lacks a centralized system for storing and managing the extensive range of research papers produced by faculty, students, and university affiliates. Without a unified repository, research papers are scattered across various personal, departmental, and digital storage locations, making it challenging for users to access, share, and reference these valuable resources.

This decentralized approach results in several issues:

1. **Limited Accessibility:** Researchers, students, and other stakeholders face difficulties accessing published works, hindering knowledge-sharing and collaboration across the university.
2. **Inefficient Retrieval:** Without a streamlined platform, the process of searching for specific papers is time-consuming and unreliable, leading to potential duplication of research efforts.
3. **Lack of Preservation and Continuity:** Valuable research may be lost or remain inaccessible due to lack of a systematic archiving process.



4. **Low Visibility:** The absence of a centralized repository limits the visibility and impact of the university's research output, both internally and externally.

1.5.1 General Objective

The objective of this study is to develop an Online Web Based Application Repository System for Research Papers in De La Salle University - Dasmariñas.

1.5.2 Specific Objectives

1. Review existing online repository platforms and systems, both locally and internationally, to identify features, functionalities, and best practices relevant to the development of the repository for DLSU-D.
2. Consult with key stakeholders within DLSU-D, including the University Research Office (URO) and the faculty of the Computer Science Department, to gather insights and requirements for the design and implementation of the online repository.
3. Conduct a needs assessment survey among the URO and the Computer Science Department faculty to identify specific preferences, challenges, and expectations regarding the access and management of research papers.



4. Develop a comprehensive framework and specifications for the design and architecture of the online repository, ensuring it meets the identified needs and aligns with institutional goals.
5. Design and prototype the user interface and functionalities of the online repository, incorporating feedback and suggestions gathered from stakeholders and usability testing sessions to enhance user experience.
6. Implement the backend infrastructure and database management system for the online repository, ensuring robustness, reliability, and scalability to accommodate a growing collection of research papers.
7. Utilize the dataset of research papers provided by the URO to populate the online repository, ensuring accurate metadata tagging and indexing for efficient search and retrieval.
8. Conduct usability testing and evaluation sessions with representative users from DLSU-D to assess the effectiveness, usability, and user satisfaction of the online repository prototype.
9. Iterate and refine the design and functionality of the online repository based on feedback and evaluation results, incorporating enhancements and addressing any identified usability issues or technical challenges.



10. Develop comprehensive documentation and training materials to support the deployment, administration, and use of the online repository within the DLSU-D academic community.

1.6 Hypothesis

The hypothesis of this study encompasses presumptions about the expected results of establishing an online repository for research publications at De La Salle University - Dasmariñas (DLSU-D). These theories aim to evaluate the anticipated influence and efficacy of the online repository in meeting the demands and specifications of the campus community. Specifically, the following hypotheses are proposed:

1. Will the online repository system improve the efficiency of accessing and retrieving research papers for students and faculty?
2. Will the system enhance the overall management of research papers by the university's research department?
3. Will the implementation of the repository lead to higher engagement and usage rates among students and future researchers?



1.7 Significance of the Study

The significance of this study lies in its potential to enhance access to research papers through an online repository, thereby benefiting various users who are expected to leverage its resources.

1. **De La Salle University - Dasmariñas** - De La Salle University - Dasmariñas can greatly benefit from the repository's extensive collection of research papers. The university can utilize this resource to access scholarly literature that supports educational objectives, informs curriculum development, and aids research projects. The repository will serve as a valuable tool for educators and students alike, providing high-quality academic content across diverse disciplines.
2. **University Research Office** - URO can utilize the online repository to showcase the scholarly work of faculty and researchers. This platform will enable them to monitor research trends, disseminate research outputs, organize research papers and evaluate the impact of ongoing research initiatives within the university.
3. **Students** - Both undergraduate and graduate students are key users of the repository. They can access research papers for various



academic purposes, thesis projects, and independent studies. The repository will serve as a critical learning resource, enabling students to explore a range of topics, gain insights from contemporary research, and cultivate critical thinking skills.

4. Professors/Teachers - Faculty members from DLSU-D can utilize the repository to stay informed about the latest research findings and trends within their fields. This platform will offer access to academic literature for research supervision, lecture preparation, and course material development. While professors and educators cannot contribute their research outputs to the repository, they can leverage the curated collection of research papers organized by the University Research Office (URO) to enhance their teaching and research activities.

5. Future Researchers - The repository will provide aspiring scholars and researchers with access to a wide array of research topics and methodologies. By reviewing previously published research articles, future researchers can better understand their fields, identify gaps in existing knowledge, and develop their research topics. Moreover, insights from this study may inform



enhancements in online repositories, contributing to the ongoing evolution and improvement of similar systems.

1.8 Scope and Delimitations

The scope of this study is centered on the design, development, and implementation of an online repository for research papers at De La Salle University - Dasmariñas (DLSU-D). This project specifically addresses URO-approved research publications, ensuring accessibility only for members of the DLSU-D academic community, including faculty, students, and administrative staff. The repository aims to enhance access to research materials, streamline the management of research outputs, and promote academic engagement within the institution. By focusing on these objectives, the study intends to create a comprehensive platform that supports the academic needs of its users.

However, the study is delimited by several factors. Firstly, the user base is restricted to the DLSU-D community, meaning individuals from other universities or the general public will not have access to the repository. Secondly, only research papers that have received approval from the University Research Office (URO) will be included, thereby maintaining high standards of quality and relevance, but excluding unpublished research or papers not sanctioned by the URO. Additionally, the study is geographically limited to the context of DLSU-D, which may affect the generalizability of findings to other institutions or



educational settings. Finally, the technical implementation will rely on specific software tools and technologies that align with the university's infrastructure, and any future enhancements beyond the initial implementation will fall outside the study's scope. By clearly defining these limitations, the study aims to set realistic expectations for its outcomes and applicability.

1.9 Definition of Terms

1. **Online Repository:** An online repository is a digital platform that allows for the storage, management, and sharing of documents and data, typically enabling users to upload, access, and retrieve various types of content. It serves as a centralized resource for academic and research materials, facilitating better organization and accessibility (Smith, 2020; Johnson & Lee, 2019).
2. **Research Papers:** Research papers are scholarly articles that present original findings or analyses based on empirical data or literature reviews. They contribute to the body of knowledge in a particular field and are often subject to peer review before publication (Jones, 2021; Brown, 2022).
3. **University Research Office (URO):** The University Research Office (URO) is the department responsible for overseeing research activities, ensuring compliance with institutional guidelines, and facilitating the dissemination of research outputs within a



university setting. The URO plays a critical role in maintaining the quality and integrity of academic research (Garcia, 2019; Smith & Taylor, 2021).

4. **Metadata:** Metadata refers to the structured information that describes and provides context for data. In the context of an online repository, metadata includes details such as the title, author, abstract, keywords, and publication date of research papers, which aid in organizing and retrieving documents efficiently (Miller, 2020; Thompson, 2022).
5. **User Interface (UI):** The user interface (UI) is the point of interaction between the user and the online repository. It encompasses the visual design, layout, and usability features that facilitate user engagement and navigation within the system (Kumar & Singh, 2021; Wilson, 2020).



CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Foreign Literature and Studies

2.1.1 Web Based Digital Repository for Scholarly Publication

Okon et al. (2020) demonstrates that web-based digital repositories significantly enhance accessibility and preservation of research outputs by offering an open-access platform that mitigates the limitations of physical storage. This advancement not only reduces congestion and wear on physical copies but also supports global academic collaboration by making research readily available to a wider audience. Additionally, the use of object-oriented analysis and design in developing these repositories ensures they are robust, scalable, and user-friendly, catering to the specific needs of institutions like De La Salle University - Dasmariñas. By implementing such systems, academic institutions can improve the efficiency of research activities and the visibility of their scholarly contributions.

2.1.2 Web-Based Online Platform for Research Study

Delera et al. (2023) aimed to develop a research repository system to securely, efficiently, and reliably manage thesis documents for undergraduate and graduate students. This system was designed to be user-friendly and comply with specified requirements. To achieve this, the researchers used the Agile model approach within the Software Development Life Cycle (SDLC) for verifying and validating



the application. To assess the system's effectiveness, a survey was administered to students and research coordinators at various universities in Negros Occidental, Philippines. The survey results indicated a strong interest in implementing the system due to its improved and well-managed data archiving solution for research digital content. In conclusion, the study successfully developed a research repository system that met the necessary requirements and demonstrated its potential for effective management of research digital content.

2.1.3 ReCOVery: A Multimodal Repository for COVID-19 News Credibility

Zhou et al. (2020). The outbreak of COVID-19, originating in Wuhan, China, in December 2019, has prompted a global emergency declaration by the World Health Organization (WHO) in January and a pandemic declaration in March 2020. Alongside this pandemic, there has been a surge of misinformation termed as an "infodemic." To address this issue, they introduce ReCOVery, a repository aimed at facilitating research on combating misinformation related to COVID-19. The repository encompasses approximately 2,000 news publishers, with 60 identified as exhibiting extreme levels of credibility, either high or low. Collecting a total of 2,029 news articles on coronavirus published from January to May 2020, along with 140,820 corresponding tweets reflecting their dissemination on Twitter, the repository provides a multimodal dataset comprising textual, visual, temporal, and network information. The authors adopt a methodology where news credibility is inherited from the credibility of the media platforms on which they



were published, striking a balance between dataset scalability and label accuracy. The repository is made available to the research community to facilitate further studies on news credibility.

2.1.4 Development of web-based repository technology for lecturer's performance monitoring

Rony, M. A., & Anggraeni, M. D. (2023). The assessment of higher education performance for accreditation purposes necessitates comprehensive documentation of lecturer performance. Existing technologies, such as FTP and Google Drive, are currently employed by the Faculty but require manual data processing, reducing their effectiveness and efficiency. To address these limitations, the authors introduce a web-based repository aimed at streamlining the monitoring and recording of lecturer performance, complete with supporting document evidence. This system enables the Faculty to efficiently manage performance data and retrieve necessary documentation for accreditation preparation. The research adopts a hybrid approach, combining Design Thinking and Scrum methodologies for the application development process.

2.1.5 Web-based proceedings article repository

Che Mohd Sakri, N. S., & Wan Ishak, W. H. (2023) This study proposes a digital repository system to manage and distribute proceedings articles, addressing the challenges of locating and retrieving specific articles due to unsystematic



management. The aim is to provide a reliable, efficient, effective, and user-friendly system for storing and retrieving these articles, thereby enhancing research productivity and quality. The project involved gathering functional requirements through requirement analysis, leading to the development of a prototype called the Web-based Digital Repository for Proceedings Articles (WDR). A field study assessed the prototype's usability, and the evaluation indicated that WDR is practical and easy to use. The system's functions for managing and retrieving proceedings articles received positive feedback from respondents.

2.2 Local Literature and Studies

2.2.1 Online Repository System for Thesis and Special Problem Manuscripts

Armacheska R. Mesa (2018) Traditional methods of storing hardbound and digital copies have resulted in problems such as duplication, loss, and difficulty in tracking borrowed manuscripts. UPMERS, a web-based information system, allows administrators to monitor and manage records of theses and special problem manuscripts submitted by graduates. This system enables the storage of digital copies in a centralized database, facilitating easier access and retrieval of scholarly works by students, faculty, and other stakeholders. The system's design follows the Rapid Application Development (RAD) model, ensuring a streamlined development process through stages of requirements planning, user design, rapid



construction, and cutover. UPMERS was evaluated using the Technology Acceptance Model (TAM) to assess user acceptance, focusing on perceived ease of use and perceived usefulness.

2.2.2 Systems development for records archiving and digital documents repository: A case study.

Gamba, M. P., Ocbian, M. M., & Gamba, M. N. (2018). In response to the growing need for efficient data access and transfer, this study focuses on developing a records archiving and document repository system for Sorsogon State College in the Philippines. The college, which comprises four campuses situated in different municipalities, faces challenges in communication and real-time updates due to their geographical separation. To address these issues, the authors developed a web-based application with a General User Interface (GUI) that can be accessed via any web-enabled computer or mobile device using standard web browsers. This system allows satellite campuses to connect to the main campus's private server through a cost-effective virtual private network (VPN) provided by their Internet service provider. The system enhances security by employing a 1024-bit Rivest-Shamir-Adleman (RSA) private/public key exchange and 256-bit Advanced Encryption Standard (AES) encryption for user credentials, along with 128-bit encryption for the contents of uploaded files to protect against unauthorized access within the intranet.



2.2.3 Development of the online repository of theses and dissertations of the University of Cebu – Graduate School Library

Dinauanao, A. M. (2019). Libraries play a crucial role in providing access to both printed and digital materials, essential in today's knowledge-based society. The University of Cebu (UC) faces challenges as its theses and dissertations from graduate students are not available online, making it difficult for library patrons to locate these resources. This study aimed to develop an online repository for these academic works. Using a combination of descriptive and quasi-experimental designs, data were collected through questionnaires, interviews, and content analysis. The resulting online repository adheres to standards for database management, search functionality, user interface, and data integrity, as verified by IT experts and end-users. The study recommends the deployment, promotion, and maintenance of the ucGSLIB to enhance resource accessibility.

2.2.4 Design and implementation of EMRC web-based portal

Mallo-Eustaquio, G. G. (2019). Design and implementation of EMRC web-based portal. This study addresses the evolving needs and demands of library patrons influenced by advancements in information and communication technology (ICT). Recognizing the growing preference for web search engines, the researcher designed and implemented an EMRC web-based portal to enhance the library's service delivery. The project aimed to provide a user-friendly,



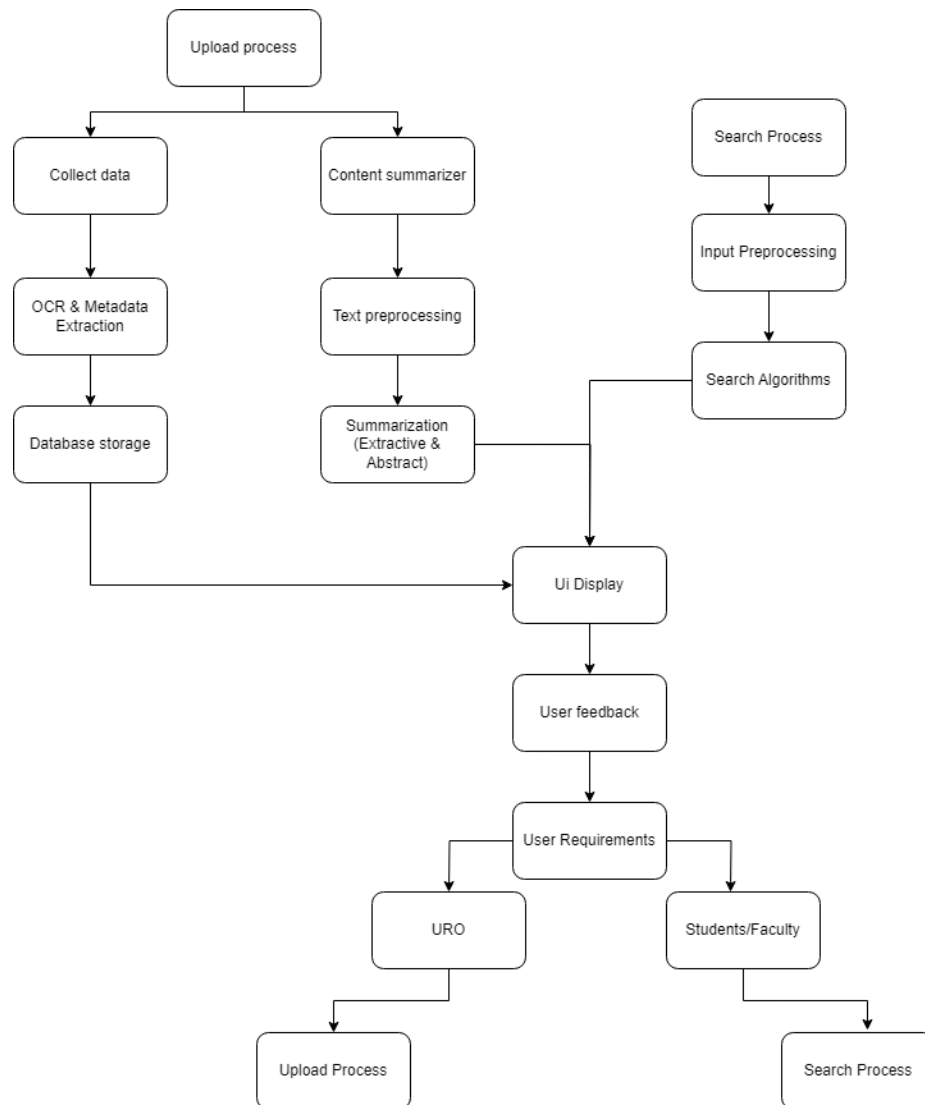
manageable, and efficient platform for online availment, reservation, and communication, ensuring accurate and effective information access. The study employed Rapid Application Development (RAD) methodology, utilizing HTML, CSS, JavaScript, PHP, and MySQL for system development. The portal's design was modeled using Unified Modeling Language (UML) and evaluated based on ISO 25010:2011 standards. The evaluation highlighted the system's strengths in security, functional suitability, maintainability, portability, usability, reliability, performance efficiency, and compatibility. The findings underscore the portal's effectiveness in improving library services and suggest areas for future enhancements. This research promotes collaboration between librarians and IT personnel and provides a basis for further developments in library services at MSEUF.



CHAPTER III

METHODOLOGY

3.1 Software Development



The figure above illustrates the research framework for the study. The system begins by gathering user requirements, which informs the development of



essential features. There are two primary workflows: the upload process and the search process.

In the upload process, the URO will upload the approved research papers, initiating a sequence where data is collected, OCR and metadata extraction is performed, and the information is stored in a centralized database. The content then goes through a summarization phase, where it is preprocessed using text tools and summarized using extractive and abstract techniques, making it easier for users to understand the content at a glance.

In the search process, students submit queries that undergo input preprocessing before being processed by search algorithms. Both the search results and any summarized content are displayed in the UI for easy access.





Figure 2. SDLC of AnimoNexus

Requirements

The first phase involves gathering detailed requirements from users to understand the necessary features and functionalities of the repository.

Design

The second phase focuses on creating wireframes, UI designs, and database schemas to outline the system's structure.

Development

The third phase is dedicated to coding and implementing the features based on the design specifications.

Testing

The fourth phase involves rigorous testing to ensure the software functions correctly and meets performance standards.

Deployment

The fifth phase is deploying the tested application to the production environment for user access.



Review

The sixth phase involves collecting user feedback and conducting reviews to identify improvements.

Acceptance

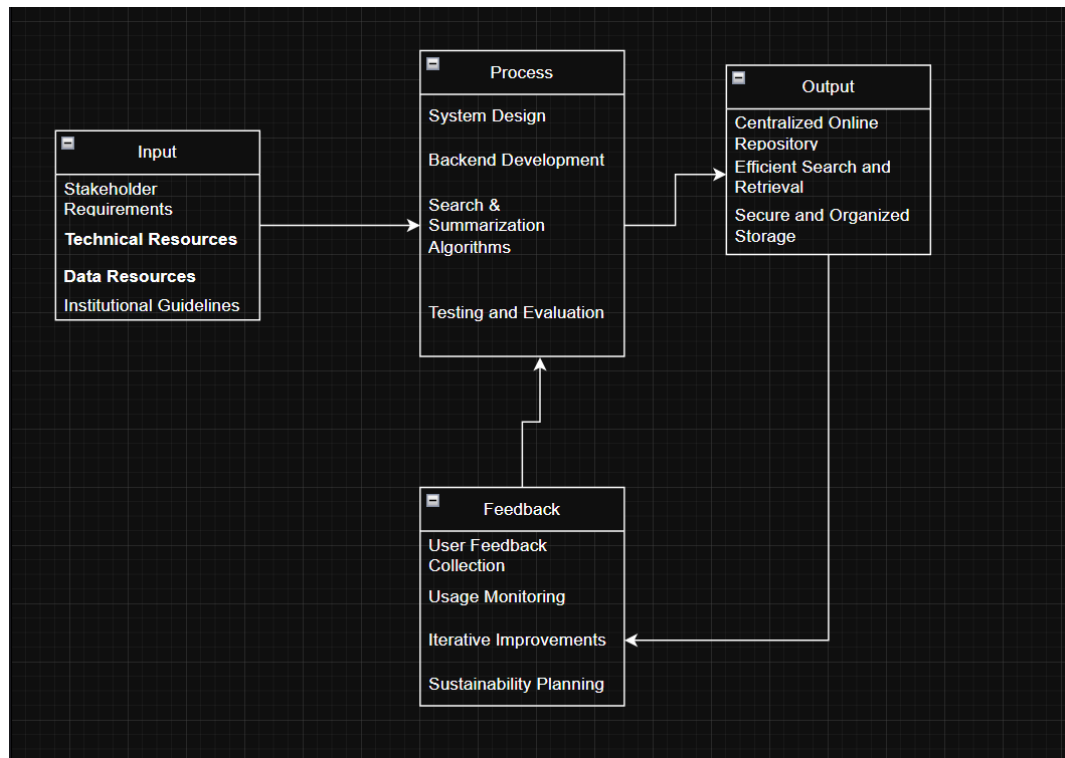
The final phase is the formal acceptance of the software, ensuring it is error-free and meets all requirements.

The development of the online research paper repository is ensured to be structured, iterative, and user-centered by adhering to these agile phases, which will result in an application that is dependable, user-friendly, and of high quality.



3.2 Conceptual Framework

Figure 3. Conceptual Framework of AnimoNexus



The framework guiding the development of AnimoNexus, a web-based repository for research papers at De La Salle University - Dasmariñas, follows an expanded Input-Process-Output-Feedback (IPOF) model. This comprehensive approach ensures not only the system's functional development but also its long-term relevance and continuous improvement.

In the Input stage, essential resources and requirements are gathered, including stakeholder feedback, URO-approved research outputs, technical tools



and platforms, and institutional guidelines on security and privacy. These inputs form the foundational elements necessary for effective system planning.

The Process stage encompasses the design and development of the system's architecture, frontend and backend components, and data integration workflows. This includes implementing features such as Optical Character Recognition (OCR), metadata extraction, and intelligent search functionalities using natural language processing. Quality assurance, usability testing, and system deployment are also conducted during this phase to ensure reliability and performance.

The Output stage produces a functional, secure, and centralized online repository. This output is characterized by enhanced access to research papers, improved user engagement, increased visibility of academic contributions, and compliance with institutional data governance standards.

Integrated into this cycle is the Feedback stage, which closes the development loop by collecting ongoing input from users through surveys, interviews, and usage analytics. This feedback is systematically analyzed to identify areas for refinement, guide future updates, and ensure the system evolves in response to user needs and institutional priorities. Regular reviews and feature enhancements based on this feedback sustain the platform's usability, scalability, and value to the university's academic and research ecosystem.



3.3 Algorithms and it's rules

3.3.1 Natural Language Processing (NLP)

Natural Language Processing (NLP) is a branch of artificial intelligence that allows computers to understand and process human language. In the development of AnimoNexus, NLP is used to improve how research papers are organized, searched, and summarized within the system.

NLP techniques help the system generate brief and relevant summaries of research papers, allowing users to understand the main ideas of a paper without reading the entire content. These techniques also support the search function by improving the accuracy of results based on the meaning of the user's query.

Before analysis, the text undergoes **preprocessing**, which prepares the content for further computation. This includes:

- **Tokenization** – Dividing the text into individual words or phrases called tokens.
- **Stop Word Removal** – Removing common words (e.g., “the”, “and”, “in”) that do not add important meaning.
- **Stemming/Lemmatization** – Reducing words to their root forms (e.g., “running” becomes “run”) to standardize the text.



Once the text is processed, the system uses techniques such as **TF-IDF** and **cosine similarity** to identify the most relevant sentences. These are used to create **extractive summaries** that highlight key information from the research paper.

Overall, NLP improves the usability of AnimoNexus by making it easier for users to find and understand research content quickly and effectively.

3.3.2 Formulas and Mathematical Operations

- **Cosine Similarity:** Cosine similarity is used to measure the similarity between two sentences or documents. This helps in selecting sentences that are most similar to the central theme or most relevant to the query.

$$\text{Cosine Similarity} = (A \cdot B) / (\|A\| * \|B\|)$$

where A and B are vectors representing sentences or documents.

The dot product $A \cdot B$ measures the similarity, and $\|A\|$ and $\|B\|$ are the magnitudes of the vectors.

Cosine similarity scores range from -1 to 1, with 1 indicating maximum similarity. Sentences with high cosine similarity to the main content are selected for the summary.



- **TF-IDF (Term Frequency-Inverse Document Frequency):**

TF-IDF is used to identify important words and sentences in a document by measuring how frequently a term appears in a document relative to its frequency in the entire corpus. This helps in scoring and ranking sentences based on their relevance.

- **Term Frequency (TF):** Measures how often a term appears in a document.

$$TF(t, d) = \frac{f_{t,d}}{N_d}$$

where $f_{t,d}$ is the frequency of term t in document d , and N_d is the total number of terms in document d .

Inverse Document Frequency (IDF): Measures how important a term is by considering how often it appears across all documents.

$$IDF(t) = \log \left(\frac{N}{|\{d \in D : t \in d\}|} \right)$$

$$IDF(t) = \log \left(\frac{N}{|\{d \in D : t \in d\}|} \right)$$



where N is the total number of documents, and $|\{d \in D : t \in d\}|$ is the number of documents containing term t .

TF-IDF Score:

- The TF-IDF score is the product of Term Frequency (TF) and Inverse Document Frequency (IDF). It evaluates the importance of a term t in a specific document d within a collection of documents. A high TF-IDF score indicates that the term is frequent in a particular document but rare across the entire document set, making it more relevant.

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t)$$

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t)$$

This score is used to rank sentences, identifying those with the highest relevance for the summary.



3.3.2 Upload Research Paper Feature

To allow the university research office (URO) to upload research papers to the online repository, ensuring they are stored securely and are easily accessible to others.

1. Algorithms used

- **Optical Character Recognition (OCR):** OCR technology is used to extract text from PDF files. Libraries such as Tesseract can be used for this purpose. OCR converts different types of documents, such as scanned paper documents or PDF files, into editable and searchable data.
- **Metadata Extraction:** Algorithms are applied to the extracted text to identify and extract metadata such as title, authors, and keywords. Named entity recognition and parsing algorithms help in identifying key information and structuring it accordingly.

3.3.3 Research Paper Search Feature

To enable users to search for research papers using various criteria such as title, author, and keywords.



1. Algorithms used

- **Substring Search:** Substring search can be used to find matches within the text of research papers. This involves searching for a substring (the search query) within a larger string (the text of the paper). This is a simple yet effective way to find occurrences of the search terms within the documents.
- **Cosine Similarity:** Used to measure the similarity between the search query and the indexed research papers. This helps in determining how closely a document matches the search query.

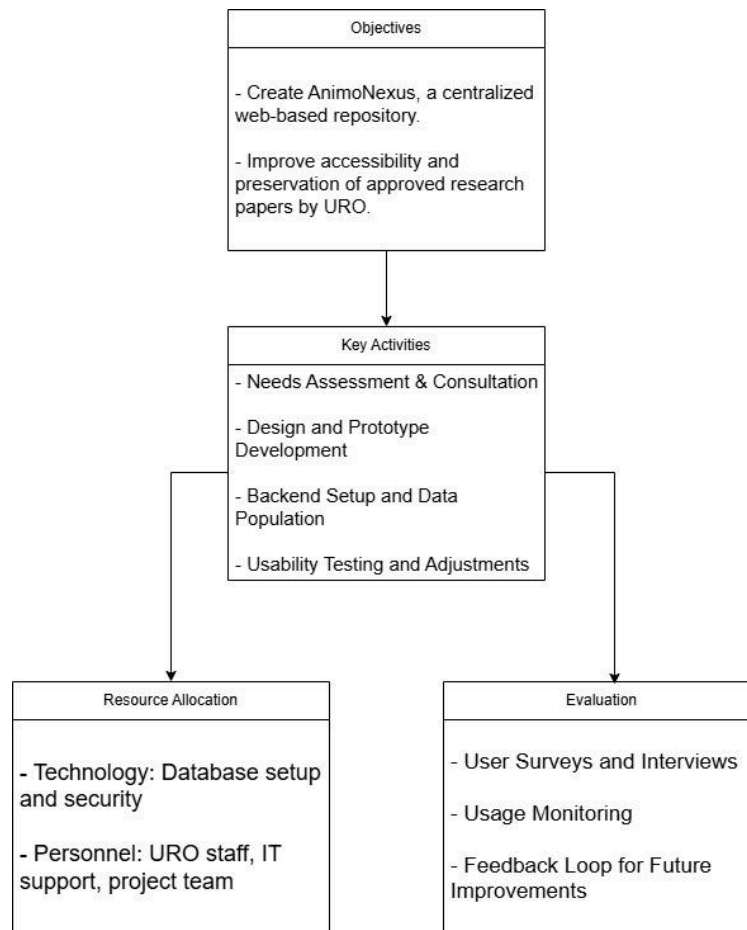
Used to measure the similarity between the search query and the indexed research papers. Cosine similarity scores range from -1 to 1, with 1 indicating maximum similarity. Papers with high cosine similarity to the search query are ranked higher in the search results.

These algorithms provide a straightforward, easy-to-implement approach to automating parts of the approval process. By focusing on substring matching, keyword categorization, and rule-based checks, the system can perform essential tasks without requiring complex code or advanced data processing models.



3.4 Operational Framework

Figure 4. Operational Framework of AnimoNexus



The figure above illustrates a streamlined workflow for developing and managing the AnimoNexus online research paper repository. It begins with **Project Objectives**, which set the goal of creating a centralized, web-based



repository to improve the accessibility, preservation, and organization of URO-approved research outputs at De La Salle University - Dasmariñas.

The **Key Activities** component outlines the core tasks necessary to achieve these objectives. This starts with a **Needs Assessment & Stakeholder Consultation** phase, where requirements are gathered from the University Research Office (URO) and key faculty members. Next, the **Design and Prototype Development** phase establishes a basic user interface and repository structure. This is followed by **Backend Setup and Data Population**, where the technical infrastructure is established, and initial URO-approved papers are added to the system. Lastly, **Usability Testing and Adjustments** are conducted, gathering user feedback to make final adjustments for improved functionality and user experience.

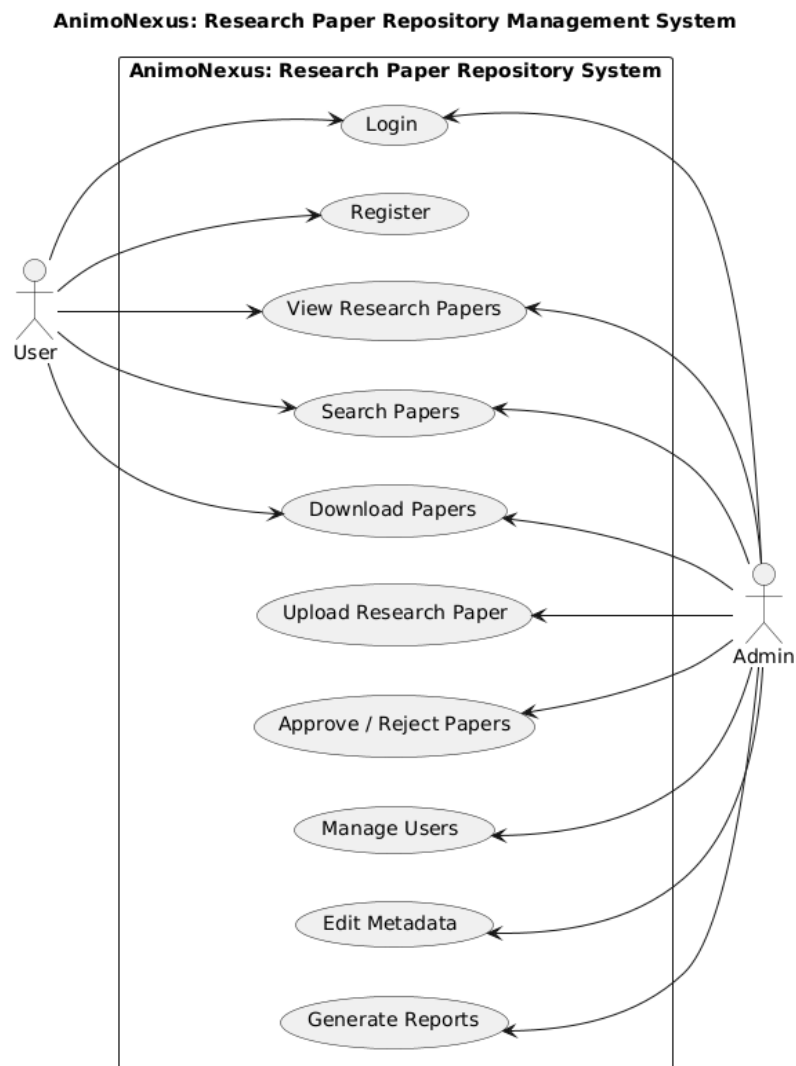
The **Resource Allocation** section defines the resources necessary for the project, including **Budget** for licensing software and basic training, **Technology** for database setup and security measures, and **Personnel** such as URO staff, and the project team responsible for development and testing.

Finally, the **Evaluation Mechanisms** ensure continuous improvement and relevance of the repository. **User Surveys and Interviews** gather direct feedback on usability and satisfaction, while **Usage Monitoring** provides insight into how frequently the repository is accessed and utilized. An **Ongoing Feedback Loop**



enables the repository team to make future enhancements based on user input, ensuring the platform remains valuable for students, faculty, and other university stakeholders.

3.5 Use Cases





The figure above illustrates the interaction between the two main types of users within the AnimoNexus: Research Paper Repository Management System Users and Admins. Users have limited access focused on exploring research content. They can register, log in, view research papers, search for specific topics, and download papers. These functions support the users' need for academic references and access to institutional research. Admins, on the other hand, have full control over the repository. In addition to the functions available to regular users, admins can upload research papers, approve or reject submissions, manage user accounts, edit paper metadata, and generate system reports. This separation of responsibilities ensures that only authorized personnel can manage the repository's content while users are free to access and consume research data. The figure demonstrates how each actor interacts with the system, maintaining structure, security, and academic integrity.

3.6 Respondents of the Study

AnimoNexus: Research Paper Repository Management System will primarily involve two key institutional groups within the university as respondents of the study. These are the University Research Office (URO) and the College of Information and Computing Sciences (CICS) Faculty. The URO serves as the central authority in managing, reviewing, and archiving all university-approved research papers. For this project, the URO will act as the admin, responsible for



uploading approved research outputs, managing user accounts, editing metadata, and generating reports.

The CICS Faculty, composed of instructors and research coordinators from the Computer Science and Information Technology departments, will serve as end users of the system. Their role includes accessing and reviewing research papers, searching the repository for reference, and downloading materials relevant to their academic and research needs.

Both the URO and CICS Faculty are located within the university campus in De La Salle University - Dasmariñas in Dasmariñas, Cavite. The estimated number of respondents for this study includes all of the personnel from the URO and members from CICS, depending on availability and consent.



CHAPTER IV RESEARCH DESIGN

CHAPTER IV

SYSTEM DESIGN AND METHODOLOGY

This chapter provides a detailed discussion on the design, development, and evaluation of *AnimoNexus*. It describes the methodology, project development process, technical specifications, and evaluation strategies used to ensure that the system meets the academic and operational needs of the University Research Office (URO) at De La Salle University - Dasmariñas.

4.1 DESIGN AND METHODOLOGY

The study employed a descriptive-developmental research design. It is descriptive because it analyzes the existing gaps in accessing research papers within the university and developmental as it involves the creation of a working web-based repository system that addresses these gaps.

A questionnaire survey was used to evaluate the system. Faculty members and URO personnel, as key users, provided feedback on system functionality, usability, and effectiveness. Convenience sampling was used for data gathering, considering the limited and specialized nature of the target participants.

After gathering the evaluation results, the data were treated using descriptive statistics to summarize user satisfaction levels. Ethical considerations were strictly followed: respondents signed an Informed Consent Form (ICF) and no sensitive or unpublished research data was included in the testing phase. Only URO-approved and publicly shared academic works were utilized. Ethical clearance for the study



was granted by the university's research ethics committee on March 2025 (See Appendix D for the signed Ethics Certificate).

4.1.1 APPLICATION INTRODUCTION

Figure 4.1.1.1

AnimoNexus Home Screen

The home page welcomes users with the university's official logo and tagline, allowing users to either browse approved research papers or login as administrators for uploading and managing papers.

Figure 4.1.1.2

AnimoNexus Login Page

The login page prompts for username and password, granting different access levels: regular users can view and search papers, while admin users can upload new research papers and manage metadata.

Figure 4.1.1.3

User Dashboard

Upon logging in, regular users are directed to a dashboard where they can browse, search, and filter research papers based on author, title, course, year, or keywords.

Figure 4.1.1.4

Admin Dashboard



Admins have access to an extended dashboard showing functionalities such as Upload Research Paper, Edit Metadata, Manage Accounts, and System Reports.

4.2 PROJECT DEVELOPMENT

The development process followed a Rapid Application Development (RAD) approach due to the need for quick prototyping and continuous stakeholder feedback. The main phases included:

- Requirements Planning: Stakeholder interviews with URO and IT faculty to gather system requirements.
- User Design: Wireframes and mockups created for user and admin workflows, validated by target users.
- Construction: Simultaneous frontend (HTML, CSS, JavaScript) and backend (PHP, MySQL) development. OCR and NLP modules were developed separately using Python.
- Cutover: The system was deployed in a test server environment and later migrated to LightNode VPS for final access.

Figure 4.2.1

System Architecture Diagram

The system is divided into modules for authentication, research paper management (upload, download, view), OCR-based text extraction, NLP-based search summarization, and administrative control panels.



4.3 DEVELOPMENT PLANNING

4.3.1 Software Specifications

The system was built using the following tools:

1. Visual Studio Code – Main code editor for frontend and backend development.
2. Google Colab – Used for training and testing the Python NLP models (summarization and metadata extraction).
3. phpMyAdmin – For database design and administration.
4. Postman – For testing RESTful API endpoints.
5. Tesseract OCR – For extracting text content from uploaded research paper PDFs.

4.3.2 Programming Languages and Libraries

- PHP – Backend web application logic.
- JavaScript/HTML/CSS – Frontend design and interactivity.
- Python – OCR processing (Tesseract) and NLP modules (using libraries like NLTK and Scikit-learn).
- MySQL – Relational database for storing user accounts, paper metadata, and system logs.



4.4 EVALUATION OF THE PROJECT

4.4.1 Respondents of the Study

A total of 50 respondents participated:

- 4 URO personnel (admins)
- 4 faculty members from the College of Science and Computer Studies (users)

4.4.2 Research Instruments and Techniques

Evaluation was conducted using a structured questionnaire based on ISO/IEC 25010:2023 Software Quality Standards, covering aspects such as functionality, usability, efficiency, reliability, and security.

4.4.3 Data Gathering

The questionnaire was administered after participants tested the system over a one-week period. Participants were asked to assess both the ease of use and the effectiveness of features like research paper upload, search functionality, and download processes.

CHAPTER V

RESULTS AND DISCUSSIONS



CHAPTER VI

CONCLUSIONS AND RECOMMENDATIONS

REFERENCES

Delera, G. M. Jr., Salao, M. J. B., & Soberano, K. T. (2023). Web-based online platform for research study. International Journal of Research in Engineering and



Science (IJRES), 11(7), 164-169. Retrieved from
<https://www.ijres.org/papers/Volume-11/Issue-7/1107164169.pdf>

Okon, R., Eleberi, E. L., & Uka, K. K. (2020). A web-based digital repository for scholarly publication. Journal of Software Engineering and Applications, 13(4), 67-75. <https://doi.org/10.4236/jsea.2020.134005>

Che Mohd Sakri, N. S., & Wan Ishak, W. H. (2023). Web-based Proceedings Article Repository. Multidisciplinary Applied Research and Innovation, 4(1), 128-132. Retrieved from
<https://penerbit.uthm.edu.my/periodicals/index.php/mari/article/view/9852>

Satina, A. M. (2018). Design and development of an online repository system for thesis and special problem manuscripts. University of the Philippines Mindanao. Retrieved from <https://www.researchgate.net/publication/353706725>

Kaminski, J., & Hooland, S. (2020). Predicting reproducibility in computer science. Proceedings of the 29th ACM International Conference on Information & Knowledge Management, 1933-1936. <https://doi.org/10.1145/3340531.3412880>



Rony, M. A., & Anggraeni, M. D. (2023). Development of web-based repository technology for lecturer's performance monitoring. JUSIKOM PRIMA (Jurnal Sistem Informasi dan Ilmu Komputer Prima), 7(2), 61-74. Retrieved from <http://jurnal.unprimdn.ac.id/index.php/JUSIKOM/article/view/3982>

Loquinario, L. P., Rufino, D. L. G., Ybut, J. C., & Pasion, A. P. (2018). The development of online repositories of researches for De La Salle Araneta University. Lagumlalang: A Refereed Journal of Interdisciplinary Synthesis, 1(1). Retrieved from <https://ejournals.ph/article.php?id=1086>

Gamba, M. P., Ocbian, M. M., & Gamba, M. N. (2020). Systems development for records archiving and digital documents repository: A case study. JPAIR Institutional Research Journal, 4(1). Retrieved from <https://ejournals.ph/article.php?id=12040>

Mallo-Eustaquio, G. G. (2019). Design and implementation of EMRC web-based portal. PAARL Research Journal, 6(1). Retrieved from <https://ejournals.ph/article.php?id=18471>